

XRPL Validator — Daily Engineering Report • 2026-04-15

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **32 commits** (xrpl-validator 32)
- 2 test / vector commits

Engineering changes

- **feat: 100k milestone push — dashboard polish + system-metrics sidecar** (670f3c7 , xrpl-validator)
- Rolling 50-sample diverged tx buffer with L prefix (ffi_engine) - New system-metrics.py sidecar on :3779 for node-resources sparklines - Dashboard component polish: metrics-grid, node-resources, speedometer simplified (-181 net) - Wire /system/metrics rewrite in next.config.js - Ignore root-level .next/ build cache leak
- **feat(xrpl-node): wire stage3 ffi overlay path behind XRPL_FFI_STAGE3 toggle** (d66055d , xrpl-validator)
Item L from the post-100k punch list — the Phase B scaffold. Default off; no behavioral change unless XRPL_FFI_STAGE3 is explicitly enabled.
- **fix(xrpl-node): incremental_sync hardcoded localhost + stage3 diagnostic** (f5d7936 , xrpl-validator) Two fixes uncovered during the Stage 3 flip on the validator host: 1. incremental_sync.rs had 11 hardcoded http://localhost:5005 URLs that bypassed the XRPL_RPC_URL env var. This was a latent bug — invisible while ws_sync kept up with tip (inc-sync never ran), but fatal on any restart that left a gap. Fix: introduce a private rpc_endpoint() helper backed by a OnceLock that honors XRPL_RPC_URL and falls back to localhost ...
- **feat(xrpl-node): recent_tx_samples rolling buffer for live per-tx visibility** (214f73d , xrpl-validator) Adds a 50-entry rolling buffer to FfiStats that captures every tx the FFI engine applies, regardless of outcome. Format: "L{ledger_seq} {tx_type}/{ter_name} {short_hash} {ms}ms mut={N}"
- **feat: stage3_enabled visibility + live TX feed in watch_engine.py** (04a9055 , xrpl-validator)
- Add stage3_enabled: bool to FfiStats, set in start_ws_sync after reading Stage3Config. Exposed via /api/engine so dashboards and scripts can show a prominent "STAGE 3: ACTIVE" banner. - watch_engine.py: add big green Stage 3 banner at the top (or gray "inactive" line when off), and a "Live TX Feed (last 15 applied)" section rendering recent_tx_samples with color-coded TER outcomes (green=tesSUCCESS, cyan=tec*, ...
- **feat(dashboard): Stage 3 banner + Live TX Feed + hash empty state fix** (00d29a1 , xrpl-validator) Three dashboard updates for Stage 3 visibility: 1. Big "★ Stage 3 Active ★" banner at the top of HeroBanner when ffi_verifier.stage3_enabled=true, with pulsing dot and accent glow. Gray "inactive · shadow-only Phase A" pill when off. Operators can tell at a glance which mode the validator is in.
- **feat(ffi): port shim to rippled 3.1.2 Application API** (15840ec , xrpl-validator) Replaces develop-era MinimalServiceRegistry with MinimalApp — a ripple::Application subclass compatible with stable rippled 3.1.2. This is the foundation for running Stage 3 against a libxrpl built from mainnet's actual running version.

- **fix(ffi): match CMakeLists.txt to rippled 3.1.2 actual dependency chain** (d4d57ff , xrpl-validator) ed25519 and secp256k1 are compiled from external/ subdirs, not conan packages. Import their pre-built .a files directly. Add all missing conan deps (gRPC, lz4, LibArchive, SOCI, SQLite3, RocksDB, nudb, abseil, Snappy, BZip2, ZLIB) that xrpld sources require.
- **fix(ffi): rename header() to info() for 3.1.2 ReadView interface** (734e59b , xrpl-validator) 3.1.2 uses info() (returning LedgerInfo const&) instead of develop's header(). LedgerInfo is a typedef for LedgerHeader so the type matches.
- **fix(ffi): use concrete OrderBookDB instead of broken NoopOrderBookDB** (0571dc5 , xrpl-validator) 3.1.2 OrderBookDB has no virtual methods - can't subclass. Use it directly; never calling setup() keeps it empty for stateless replay.
- **fix(ffi): correct proto include path to pb-xrpl.libpb, drop stale libxrpl dir** (9504e92 , xrpl-validator)
- **fix(ffi): add proto subdirectory include path for relative pb.h includes** (ccbe073 , xrpl-validator)
- **fix(ffi): exclude Application.cpp (defines ApplicationImp, we use MinimalApp)** (9bfd38b , xrpl-validator)
- **fix(ffi): selective xrpld glob - only tx engine + dependencies** (1ee7aac , xrpl-validator) Exclude overlay, peerfinder, consensus, perflog, rpc directories that use incompatible boost::asio executor APIs. Include only the subsystems apply()/preflight() actually touches.
- **fix(ffi): exclude xrpld infrastructure files that pull overlay/consensus deps** (4414f2c , xrpl-validator) LedgerMaster, ValidatorList, NetworkOPs, PathRequests etc reference overlay Message/InfoSub symbols. Exclude them - not needed for apply.
- **fix(ffi): provide Application ctor, exclude BookListeners/Ledger, add gc-sections** (6f7abb1 , xrpl-validator) Application::Application() was defined only in Application.cpp which we exclude. Provide it in MinimalApp.cpp. Exclude BookListeners.cpp (InfoSub deps) and Ledger.cpp (LedgerToJson deps). gc-sections strips remaining dead code that references overlay/consensus symbols.
- **fix(ffi): provide isFlagLedger definition (excluded with Ledger.cpp)** (6633d5c , xrpl-validator) Simple seq % 256 check needed by Change.cpp pseudo-tx handler.
- **fix(ffi): add ed25519/secp256k1 link paths in build.rs** (8386d98 , xrpl-validator) These are built from rippled's external/ dirs, not conan packages. Need explicit link search paths + lib names for Rust linker.
- **fix(ffi): exclude conan OpenSSL from Rust link (version conflict with system 3.x)** (1697fad , xrpl-validator) Conan provides OpenSSL 1.1.x static libs. The Rust openssl crate links system OpenSSL 3.x dynamically. Mixing them causes missing 3.x-only symbols. Let Rust handle its own OpenSSL linking.
- **fix(ffi): skip entire OpenSSL conan package in build.rs** (fc1947d , xrpl-validator) Excluding just lib names wasn't enough - the conan OpenSSL search paths cause the linker to find 1.1.x .a files before system 3.x .so. Skip the OpenSSL cmake data file entirely.
- **fix(ffi): use FFI drops_destroyed for fee tracking instead of byte scanner** (2731504 , xrpl-validator) extract_fee_drops was a byte-pattern scanner that matched 0x68 0x4_ in raw tx blobs. False positives inside signatures produced absurd fee values (10¹⁹ drops). Use the FFI apply result's drops_destroyed which comes from libxrpl's actual transaction processing.
- **fix(dashboard): remove fake validator agreement percentages** (3bc9c5a , xrpl-validator) The consensus grid had hardcoded fake lower percentages for V14, V28, V32, V34 etc. that made them appear non-green. Replace with real active/inactive state based on actual agreement count from backend.
- **fix(dashboard): update validation target to 50K, rename to Validation Window** (6d25337 , xrpl-validator)

- **fix(sync): abort write batch when any SLE fetch fails** (e9e3fbc , xrpl-validator) Previously, process_ledger wrote all successfully-fetched SLEs to state.rocks even when some fetches failed. The missing keys silently corrupted the state, causing hash mismatches ledgers later when transactions touched the missing entries.
- **fix(dashboard): use shadow_hash_attempted for ledger count and progress bar** (911bd91 , xrpl-validator) The validation window tracks verified shadow hash comparisons, not raw FFI engine ledger count. Renamed label to "Verified".
- **feat(ffi): add persistent shadow hash mismatch logger** (5c90319 , xrpl-validator) Writes to logs/shadow_mismatches.jsonl (or XRPL_FFI_SHADOW_LOG env). Each entry records timestamp, ledger seq, our hash, network hash, and overlay key count. Same JSONL pattern as divergences.jsonl.
- **fix(ffi): get ledger seq from stats for shadow mismatch logger** (9540234 , xrpl-validator)

Tests and campaigns

- **test(xrpl-ffi): TicketCreate harness tests for the post-100k divergence hunt** (a62144b , xrpl-validator)
- **test(xrpl-node): apply_ledger_in_order reproducer for TicketCreate cascade** (448ba95 , xrpl-validator)

Docs and operations

- **chore: update libxrpl version references from 3.2.0-b0 to 3.1.2** (360cae9 , xrpl-validator)
- **docs: why libxrpl was downgraded from develop (3.2.0-b0) to stable (3.1.2)** (879f389 , xrpl-validator)
- **diag(ffi): dump FFI vs AffectedNodes overlay diff on shadow mismatch** (5be9359 , xrpl-validator)

Appendix — commits

Time	Repo	Commit	Subject
07:21	xrpl-validator	670f3c7	feat: 100k milestone push — dashboard polish + system-metrics sidecar
07:41	xrpl-validator	d66055d	feat(xrpl-node): wire stage3 ffi overlay path behind XRPL_FFI_STAGE3 toggle
08:23	xrpl-validator	a62144b	test(xrpl-ffi): TicketCreate harness tests for the post-100k divergence hunt
09:14	xrpl-validator	448ba95	test(xrpl-node): apply_ledger_in_order reproducer for TicketCreate cascade
19:53	xrpl-validator	f5d7936	fix(xrpl-node): incremental_sync hardcoded localhost + stage3 diagnostic
20:06	xrpl-validator	214f73d	feat(xrpl-node): recent_tx_samples rolling buffer for live per-tx visibility
20:12	xrpl-validator	04a9055	feat: stage3_enabled visibility + live TX feed in watch_engine.py
20:28	xrpl-validator	00d29a1	feat(dashboard): Stage 3 banner + Live TX Feed + hash empty state fix
20:54	xrpl-validator	15840ec	feat(ffi): port shim to rippled 3.1.2 Application API
21:17	xrpl-validator	d4d57ff	fix(ffi): match CMakeLists.txt to rippled 3.1.2 actual dependency chain
21:19	xrpl-validator	734e59b	fix(ffi): rename header() to info() for 3.1.2 ReadView interface
21:21	xrpl-validator	0571dc5	fix(ffi): use concrete OrderBookDB instead of broken NoopOrderBookDB
21:23	xrpl-validator	9504e92	fix(ffi): correct proto include path to pb-xrpl.libpb, drop stale libxrpl dir
21:24	xrpl-validator	ccbe073	fix(ffi): add proto subdirectory include path for relative pb.h includes
21:25	xrpl-validator	9bfd38b	fix(ffi): exclude Application.cpp (defines ApplicationImp, we use MinimalApp)
21:29	xrpl-validator	1ee7aac	fix(ffi): selective xrpld glob - only tx engine + dependencies
21:31	xrpl-validator	4414f2c	fix(ffi): exclude xrpld infrastructure files that pull overlay/consensus deps

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21:33	xrpl-validator	6f7abb1	fix(ffl): provide Application ctor, exclude BookListeners/Ledger, add gc-sections
21:36	xrpl-validator	6633d5c	fix(ffl): provide isFlagLedger definition (excluded with Ledger.cpp)
21:38	xrpl-validator	8386d98	fix(ffl): add ed25519/secp256k1 link paths in build.rs
21:39	xrpl-validator	1697fad	fix(ffl): exclude conan OpenSSL from Rust link (version conflict with system 3.x)
21:40	xrpl-validator	fc1947d	fix(ffl): skip entire OpenSSL conan package in build.rs
21:53	xrpl-validator	2731504	fix(ffl): use FFI drops_destroyed for fee tracking instead of byte scanner
21:59	xrpl-validator	360cae9	chore: update libxrpl version references from 3.2.0-b0 to 3.1.2
22:01	xrpl-validator	879f389	docs: why libxrpl was downgraded from develop (3.2.0-b0) to stable (3.1.2)
22:04	xrpl-validator	3bc9c5a	fix(dashboard): remove fake validator agreement percentages
22:07	xrpl-validator	6d25337	fix(dashboard): update validation target to 50K, rename to Validation Window
22:13	xrpl-validator	e9e3fbc	fix(sync): abort write batch when any SLE fetch fails
22:19	xrpl-validator	911bd91	fix(dashboard): use shadow_hash_attempted for ledger count and progress bar
22:22	xrpl-validator	5c90319	feat(ffl): add persistent shadow hash mismatch logger
22:24	xrpl-validator	9540234	fix(ffl): get ledger seq from stats for shadow mismatch logger
22:49	xrpl-validator	5be9359	diag(ffl): dump FFI vs AffectedNodes overlay diff on shadow mismatch